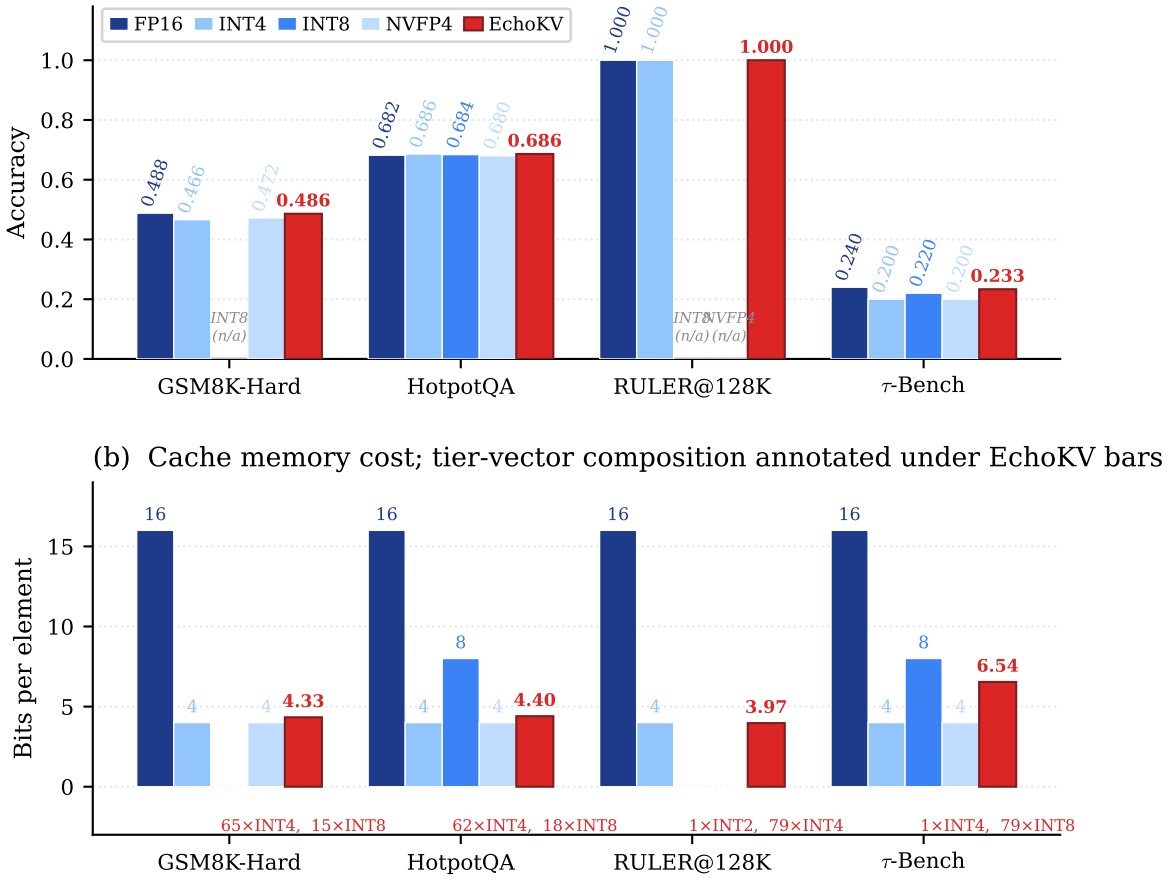
(a) Task accuracy (paired protocol; same prompts across cells)
FP16
INT4
INT8
NVFP4
EchoKV
Accuracy
0.0
0.2
0.4
0.6
0.8
1.0
0.488
0.466
INT8 (n/a)
0.472
0.486
0.682
0.686
0.684
0.680
0.686
1.000
1.000
INT8 (n/a)
NVFP4 (n/a)
1.000
0.240
0.200
0.220
0.200
0.233
GSM8K-Hard
HotpotQA
RULER@128K
τ-Bench
(b) Cache memory cost; tier-vector composition annotated under EchoKV bars
Bits per element
0
5
10
15
16
4
4
4.33
16
4
8
4
4.40
16
4
3.97
16
4
8
4
6.54
65×INT4, 15×INT8
62×INT4, 18×INT8
1×INT2, 79×INT4
1×INT4, 79×INT8
GSM8K-Hard
HotpotQA
RULER@128K
τ-Bench